Outlook

Final-year project feedback

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Dear JOE CHENG (25044810),

Please find below comments and feedback from the examiners for your final-year project undertaken in 2025-2026.

INTRODUCTION
Category description:
- clear aims and/or hypothesis
- comprehension of task
Provides a context for the work; Clearly defines the problem to be addressed; Motivates the importance of the work; Establishes the relevance in the project area.
Marking criteria description:
Defines the project area to a good standard within the subject, with some detail or sense of the wider scope of the work.
Marker's comments:
- The abstract posits that "instrument boundaries are often difficult to resolve because tools may be thin, reflective, blurred by motion, or partly hidden by tissue". However, the physical characteristics of tools do not inherently confound deep learning segmentation models, provided that sufficient and representative training data are available. The primary challenges in the field are more accurately characterised as data imbalance and data scarcity.
- While the text appropriately cites literature detailing downstream applications, the inclusion of a visual figure and an in-depth operational explanation of at least one specific clinical use case would have substantially strengthened the project's motivational framing.
- The assertion regarding whether improvements continue "once a model has already been trained on a moderate dataset" lacks quantitative precision. The dissertation should establish a rigorous definition or evaluation metric to classify what constitutes a "moderate" versus an extensive dataset in this context.
- The formulated research questions demonstrate limited methodological ambition. A more rigorous approach would have involved stratifying the data, conducting a systematic failure analysis of the baseline model, and subsequently proposing algorithmic enhancements to mitigate the identified limitations. Restricting the project scope to the development of a single baseline yields a constrained academic contribution.
- The Introduction exhibits structural redundancy, as several statements outlining the study's contributions merely reiterate points established in preceding paragraphs.
LITERATURE REVIEW
Category description:
- work described and cited
The discussion is structured and coherent; Key technical issues are summarised; Published work has been related to the project work; Literature sources have been critically analysed; The literature has been comprehensively covered.
Marking criteria description:
Reviews the background to project. Identifies the main issues, some of which are then analysed and developed.
Marker's comments:

- The section provides an adequate survey of contemporary surgical datasets utilised in state-of-the-art research.
- The inclusion of a foundational taxonomy outlining the distinct paradigms within image segmentation (e.g., semantic, instance, and panoptic) would have provided necessary theoretical context.
- Section 2.1.4, "Dataset Comparison and Selection Rationale" lacks contextual relevance, given that the LEMON dataset was pre-assigned by the supervisor. The section omits a critical comparative discussion justifying why specific datasets were deemed appropriate or inappropriate for model evaluation.
- As the experimental scope is restricted to a single baseline method, the literature review represents a missed opportunity to demonstrate a profound theoretical comprehension of that specific model's architectural intricacies.
- The dissertation enumerates multiple loss functions and proposes an ablation study. However, it lacks a data-driven rationale mathematically justifying the selection of these specific functions for ablation. Furthermore, for establishing a baseline, deploying a universally adopted loss function from current state-of-the-art literature would be methodologically sounder.

SPECIFICATION AND DESIGN

Category description:
- design technique
- rationale for approach
- difficulty and originality

Requirements have been described adequately; A rationale for the design approach is given; Alternative designs have been considered.

Marking criteria description:

Explains the rationale behind the design for the artefact comparing some advanced alternatives. Describes the design for the artefact in enough detail for reader to appreciate all significant points. Reveals some competence in the application of specialised knowledge, but lacks insight and fluency.

Marker's comments:
- The utilisation of the validation Dice coefficient to govern checkpoint retention and early stopping across all experiments (including those trained exclusively on BCE) creates a fundamental optimisation misalignment. As acknowledged in the text, validation loss and validation Dice do not strictly correlate. The network is mathematically instructed to optimise pixel-wise log-likelihood but is evaluated for check-pointing based on region overlap, which risks discarding better-calibrated probability models.
- The imposition of a strict maximum limit of 200 training epochs lacks theoretical justification. Training a deep neural network constitutes a continuous optimisation problem that should ideally proceed until convergence criteria are met, rather than terminating at an arbitrary threshold.
- Given that U-Net is a fully convolutional architecture capable of accommodating variable input dimensions, the constraint requiring square images and a fixed spatial resolution remains unjustified and warrants explicit mathematical or computational rationale.
- Equation 3.7 lacks mathematical rigour. The empirical risk minimisation problem must be formally articulated, explicitly denoting the expectation or average computed over the training distribution.
- Rather than employing random sampling to select 4,000 images from the 400 source videos, the methodology would have benefited from an active learning strategy to systematically identify the most informative frames for annotation.
- The evaluation of loss weights appears entirely heuristic. The dissertation fails to employ systematic hyperparameter optimisation to mathematically justify the selection of these specific coefficients.
- The application of a fixed decision threshold to convert foreground probabilities into binary masks is methodologically flawed when evaluating asymmetric objective functions, such as the Tversky loss. As such functions intentionally skew the predicted probability distribution, applying a static threshold conflates genuine algorithmic enhancement with a mere shift along

the Precision-Recall curve. The decision threshold must be mathematically optimised against the validation set.
IMPLEMENTATION/TECHNICAL ACHIEVEMENT
Category description:
- testing/assessment metrics
- artefact functionality
- results
The implemented product is sophisticated; Adequate and effective testing has been performed; Implementation choices have been justified; The results of testing have been analysed and discussed; Quality of the developed analytical models/simulation results/software/hardware.
Marking criteria description:
The student has solved a well-defined, moderately low-risk problem. The relevant challenges were mostly solved and some unexpected difficulties were resolved to obtain a worthwhile result. Testing and/or verification are poorly conducted.
Marker's comments:
- The decision to downsample inputs to 304x304 pixels is not adequately defended. U-Net is fully convolutional, unnecessary downsampling induces an irreversible loss of high-frequency spatial information critical for delineating fine structures, such as instrument tips. The dissertation provides no theoretical, mathematical, or hardware-constrained justification for this highly unconventional dimension.
- It is unclear whether the various proposed data augmentations were systematically ablated, as Section 3.5 omits this discussion. All architectural and preprocessing design choices must be empirically justified and validated through rigorous experimental ablation.
- An expanded, mathematically grounded explanation of the specific operations and layer configurations within the U-Net architecture would have better demonstrated the student's theoretical mastery of the model.
- The rigorous mathematical formalisation of the compared loss functions is a commendable aspect of the technical presentation.
- The generalised weighted loss formulation includes redundant objective terms, and the text lacks a rigorous explanation regarding the mathematical strategy for combining these competing loss gradients.
- The statement indicating that the patience value was set to either 15 or 8 epochs lacks sufficient methodological transparency. The precise experimental conditions dictating the selection of either value must be explicitly defined.
- By partitioning the dataset at the "individual-image level" and explicitly acknowledging that visually related frames from the same surgical sequence may appear in different subsets, the methodology introduces catastrophic temporal data leakage. The test set ceases to be statistically independent, rendering the reported out-of-distribution generalisation metrics scientifically invalid.
EVALUATION
Category description:
- interpretation of results
- evaluation of approach
- wider implications of work
Material has been compared and contrasted; Comparison of achievements with requirements; Awareness of potential extensions; Reflective and justified conclusions have been drawn.
Marking criteria description:
Evaluates the artefact to a basic standard.
Marker's comments:
- Empirical results are reported exclusively as means and standard deviations across five folds. The complete absence of formal statistical significance testing makes it impossible to ascertain whether the observed performance disparities among the evaluated loss formulations, optimisers, or resolutions represent genuine algorithmic improvements or merely stochastic variance.
- The analysis of the sample-size learning curve relies entirely on subjective visual inspection to

conclude that the curve had not clearly plateaued. To ensure scientific rigour, a formal mathematical fit (such as an inverse power-law model) should have been applied to objectively calculate the saturation asymptote and accurately project the data volume requisite for convergence.
- The two spatial resolutions evaluated in the ablation study are too proximate to reveal scaling behaviours effectively, testing a significantly larger interval, such as doubling the baseline resolution, would have yielded more conclusive insights.
GENERAL SCHOLARSHIP
Category description:
- style, grammar, layout
- future work
The writing is clear, coherent, repetition-free; Ability to express well-formed arguments; Well structured report; Free of grammatical and typographical errors; A full set of references is included; The report size limit of 25,000 words (ca 75 pages) has been met.
Marking criteria description:
Satisfactory standards of presentation.
Coherent structure to the report. Grammatical style, possibly with some errors. Use of references, diagrams, figures etc. Consistent formatting and correct use of bibliographical conventions. States some conclusions that can be drawn from the work done.
Marker's comments:
- The primary methodological diagram contains significant typographical errors, misspelling "preparation" as "prepartion" and erroneously describing the final layer as producing a "logic output" rather than a "logit output." Such oversights suggest a lack of meticulous proofreading and potentially reflect a fundamental misunderstanding of the network's terminal activation space.
LEGAL, SOCIAL, ETHICAL AND PROFESSIONAL...
Category description:
Assessment of the student's awareness, understanding and critical appreciation of the legal, social, ethical and professional issues of relevance to their project work.
Marking criteria description:
Indicates a few professional standards, codes of conduct and legislation of relevance to the project conducted. It includes a discussion of a few relevant aspects such as: effects on the public well-being, computer security, environmental and sustainability aspects, economic and commercial factors, globalisation, accessibility, Intellectual Property and related issues, software trustworthiness. Their link to the project is not clear, or well articulated.
Marker's comments:
- Although the dissertation notes that the utilisation of the LEMON dataset is subject to "non-commercial research conditions", it neglects to explicitly cite the specific CC-BY-NC-4.0 license governing the data usage rights.
- The text lacks a critical discussion regarding the systemic geographical and selection biases inherent in compiling surgical video datasets curated from publicly accessible internet repositories.
- The analysis entirely omits the statutory frameworks that would govern the hypothetical clinical deployment of this technology. The dissertation misses a critical opportunity to explicitly contextualise the algorithm within the UK MHRA Software and AI as a Medical Device (SaMD) guidelines or the EU Medical Device Regulation (EU MDR).

You should have seen your project mark already, but if you haven't, please refer to Student Records.

Kind regards,
NMES Postmaster

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